

# scripts/autonomous-qa/run-nav-challenges.sh — User Guide

**Last verified:** 2026-07-26 (LVA-018 script-docs backfill cycle)

**Inheritance:** HelixConstitution §11.4.18 (script docs); Lava §6.AK (cycle coverage), §6.Z (device evidence), §6.AH (containerized emulator)

**Classification:** project-specific

## Overview

Boots ONE containerized-KVM emulator (via `lib-emulator.sh` — the same path the autonomous-QA matrix uses) and runs the §6.AK cycle-coverage-map's supporting nav Challenges (C24 / C46 / C55) against the already-built debug APK, then tears the emulator down. Captures the JUnit XML + per-test verdict so the §6.Z device evidence can record each Challenge's pass/fail with a real device.

## Usage

```
scripts/autonomous-qa/run-nav-challenges.sh <evidence-dir> <fqns>[  
    <fqns>...]
```

*# Example: the nav-Challenge trio*

```
scripts/autonomous-qa/run-nav-challenges.sh \  
    .lava-ci-evidence/nav-challenges/$(date +%F) \  
    lava.app.challenges.Challenge24TopicNavigationTest \  
    lava.app.challenges.Challenge46SearchNavigationTest \  
    lava.app.challenges.Challenge55ForumNavigationTest
```

The debug APK must already exist at `app/build/outputs/apk/debug/app-debug.apk` (missing → exit 2). Class filtering happens via the instrumentation runner argument (`-Pandroid.testInstrumentationRunnerArguments.class=<comma-separated FQNs>`), NOT gradle's `--tests` — mirroring `run-iteration.sh`.

## Harness-side setup (why the extra steps)

- **Fresh install** (uninstall + install -r) — clean state.
- **POST\_NOTIFICATIONS pre-granted** — otherwise LeakCanary (debug-only) launches its RequestPermissionActivity mid-test → the system GrantPermissionsActivity pops OVER MainActivity → the next Compose query throws "No compose hierarchies found". (API 33+ only; ignored on older.)
- **Keyguard dismissed + screen woken BEFORE instrumentation** — a cold-booted emulator can present the lockscreen, so the first Challenge's MainActivity has no reachable compose hierarchy (the 1077 keyguard incident; run-genymotion-challenges.sh applies the same fix).
- **Logcat captured** to <evidence-dir>/raw/logcat.txt for the whole run.

## Evidence + exit

```
<evidence-dir>/
*.xml                # curated JUnit results (copied from
androidTest-results/connected)
raw/logcat.txt
raw/gradle-nav.log
```

Exit code is the gradle connectedDebugAndroidTest return code (0 = all Challenges passed); 2 = setup error (missing APK/emulator). The emulator is always torn down (EXIT trap).

## Companion files

- scripts/autonomous-qa/lib-emulator.sh (+ its doc) — the containerized emulator lifecycle
- scripts/run-genymotion-challenges.sh / scripts/run-challenge-matrix.sh (+ their docs) — the sibling Challenge runners
- scripts/autonomous-qa/aggregate-evidence.sh — the \$6.AK cycle-coverage rollup these results feed